

MSSP Report

1. Project Information

Date	9/16/2025
Project name	Project_20250916_Gabriel Mariño
Project address	
Country	Argentina
State	
City	Buenos Aires
Client name	
Client address	
Designed by	Ing. Gabriel Liguori
Reference	New Project
Revision	
Altitude(m)	0
Indoor DB temperature in cooling(°C)	24
Indoor WB temperature in cooling(°C)	17
Outdoor DB temperature in cooling(°C)	31.1
Outdoor WB temperature in cooling(°C)	23.2
Indoor DB temperature in heating(°C)	21
Indoor WB temperature in heating(°C)	14
Outdoor DB temperature in heating(°C)	4
Outdoor WB temperature in heating(°C)	0

2. Overall Material List

2.1 Equipment List

Model	Quantity	Description
MDV-V60WDHN1(AtB)	1	Atom B (Heat Pump)
MDV-D18T2/N1-DA5(At)	4	Medium Static Pressure Duct
DXFQT2-02	3	Branch header
WDC-86E/KD	4	2nd generation wired controller
RM12F	4	Remote controller

2.2 Field Providing List

2.2.1 Refrigerant Piping Materials

Model	Quantity	Unit	Description
Φ9.52	22.1	m	Copper pipe
Φ15.9	18.4	m	Copper pipe
Φ19.1	3.7	m	Copper pipe
Insulation casing for piping			All refrigerant piping and branch joints should be completely insulated.

Recommended insulation casing thickness:

Piping size	Thickness	
	Humidity<80%RH	Humidity≥80%RH
Φ6.35~Φ38.1mm	≥15mm	≥20mm
Φ41.3~Φ38.1mm	≥20mm	≥25mm

2.2.2 Refrigerant charge

System name	Model	Quantity	Unit	Description
System1	R410A	1.35	kg	Extra Refrigerant Added

2.2.3 Electrical cables

Type	Size	Length
Power supply cable	Select based on MCA of each unit	According to the actual system design
Communication cable	PQE:0.75mm2 three-core shielded cable	According to the actual system design

3. Overall Electrical Characteristics

Model	Quantity	Power supply	MCA(A)	MFA(A)
MDV-V60WDHN1(AtB)	1	220-240V-50/60Hz	40,00	40
MDV-D18T2/N1-DA5(At)	4	220-240V-50Hz	0,31	15

Notes:

1. MCA: Minimum Circuit Amps. MCA is used to select wire size.The value in above table is for one unit.

2. MFA: Maximum Fuse Amps. MFA is used to select overcurrent circuit breakers and residual-current circuit breakers. The value in above table is for one unit.

4. System1

4.1 BOM List (System1)

Model	Quantity	Unit	Description
MDV-V60WDHN1(AtB)	1		Atom B (Heat Pump)
MDV-D18T2/N1-DA5(At)	4		Medium Static Pressure Duct
DXFQT2-02	3		Branch header
WDC-86E/KD	4		2nd generation wired controller
RM12F	4		Remote controller
R410A	1.35	kg	Extra Refrigerant Added
Φ9.52	22.1	m	Copper pipe
Φ15.9	18.4	m	Copper pipe
Φ19.1	3.7	m	Copper pipe

4.2 Indoor Unit Details (System1)

4.2.1 Indoor Unit Details Table

IDU Name	Model	Weight(kg)	Dimension(WxHxD)(mm)	Power supply	MCA(A)	MFA(A)
IDU1	MDV-D18T2/N1-DA5(At)	22	997*210*500	220-240V-50Hz	0,31	15
IDU2	MDV-D18T2/N1-DA5(At)	22	997*210*500	220-240V-50Hz	0,31	15
IDU3	MDV-D18T2/N1-DA5(At)	22	997*210*500	220-240V-50Hz	0,31	15
IDU4	MDV-D18T2/N1-DA5(At)	22	997*210*500	220-240V-50Hz	0,31	15

IDU Name	Model	Tmp-C(°C)	RTC(kW)	ATC(kW)	RSC(kW)	ASC(kW)	PI-C(W)	Tmp-H(°C)	RHC(kW)	AHC(kW)	PI-H(W)
IDU1	MDV-D18T2/N1-DA5(At)	24,0/17,0		4,56		4,01	103	21		4,99	103
IDU2	MDV-D18T2/N1-DA5(At)	24,0/17,0		4,56		4,01	103	21		4,99	103
IDU3	MDV-D18T2/N1-DA5(At)	24,0/17,0		4,56		4,01	103	21		4,99	103
IDU4	MDV-D18T2/N1-DA5(At)	24,0/17,0		4,53		3,99	103	21		4,99	103

IDU Name	Model	Airflow(m³/h)	Sound-Pr dB(A)	ESP(Pa)
IDU1	MDV-D18T2/N1-DA5(At)	800[H]	36[H]	10(0~30)
IDU2	MDV-D18T2/N1-DA5(At)	800[H]	36[H]	10(0~30)
IDU3	MDV-D18T2/N1-DA5(At)	800[H]	36[H]	10(0~30)
IDU4	MDV-D18T2/N1-DA5(At)	800[H]	36[H]	10(0~30)

IDU Name	Model	Piping Length to 1st Y Joint(m)
IDU1	MDV-D18T2/N1-DA5(At)	2,50
IDU2	MDV-D18T2/N1-DA5(At)	2,70
IDU3	MDV-D18T2/N1-DA5(At)	5,40
IDU4	MDV-D18T2/N1-DA5(At)	14,60

4.2.2 Table of Abbreviations

Abbreviation code	Description
Tmp-C	Indoor temperature in cooling (Dry bulb temp. / Wet bulb temp. / RH)
RTC	Required total cooling capacity
ATC	Available total cooling capacity
RSC	Required sensible cooling capacity
ASC	Available sensible cooling capacity
Tmp-H	Indoor temperature in heating (Dry bulb temp.)
RHC	Required heating capacity
AHC	Available heating capacity
Tdis-H	Indoor unit discharge air temperature in heating
Airflow	Indoor unit airflow (High/Medium/Low)
ESP	External static pressure
Sound-Pr	Sound pressure level (High/Medium/Low)
Sound-Po	Sound power level (High/Medium/Low)
MCA	Minimum Circuit Amps
MFA	Maximum Fuse Amps
PI-C	Power input in cooling
PI-H	Power input in heating

Power supply	Power supply
Dimension(WxHxD)	Net Dimension (WxHxD) mm
Weight	Weight

4.3 Outdoor Unit Details (System1)

4.3.1 Outdoor Unit Details Table

Model		MDV-V60WDHN1(AtB)
Module		MDV-V60WDHN1(AtB)
Tmp-C	°C	31.1
RTC	kW	
ATC	kW	18,22
PI-C	kW	5,66
EER		3,22
Tmp-H	°C/°C	4/0
RHC	kW	
AHC	kW	19,97
PI-H	kW	5,96
COP		3,35
CR		128,0
Airflow	m ³ /h	5300
Sound-Pr		57
Sound-Po		
Bas-Refr	kg	4,60
Ex-Refr(ODU)	kg	0,00
Ex-Refr(Piping)	kg	1,35
TCO2 eq.		12,41
MCA	A	40
MFA	A	40
Power supply	V/ph/Hz	220-240V-50/60Hz
Dimension (WxHxD)	mm	1040*865*523
Weight	kg	91

4.3.2 Table of Abbreviations

Abbreviation code	Description
Tmp-C	Outdoor conditions in cooling (Dry bulb temp.)
RTC	Required cooling capacity
ATC	Available cooling capacity
PI-C	Power input in cooling
EER	EER
Tmp-H	Indoor conditions in heating (Dry bulb temp. / Wet bulb temp. / RH)
RHC	Required heating capacity
AHC	Available heating capacity
PI-H	Power input in heating
COP	COP
CR	Combination ratio
Airflow	Outdoor unit airflow
Sound-Pr	Sound pressure level
Sound-Po	Sound power level
Bas-Refr	Standard factory refrigerant charge
Ex-Refr(ODU)	Extra refrigerant charge for outdoor unit
Ex-Refr(Piping)	Extra refrigerant charge for liquid piping
TCO2 eq.	Tonnes of CO2 equivalent
MCA	Minimum Circuit Amps
MFA	Maximum Fuse Amps
Power supply	Power supply
Dimension (WxHxD)	Net Dimension (WxHxD) mm
Weight	Weight

4.4 Piping Limitations (System1)

4.4.1 Piping Limitations

Item	Capability	Actual Value
Total piping length	130,00(m)	23,60(m)
Longest actual length	60,00(m)	16,60(m)
Longest equivalent length	70,00(m)	18,10(m)
Longest equivalent length after first branch	20,00(m)	14,60(m)
Indoor unit to nearest branch length	15,00(m)	10,70(m)
Length difference between longest and shortest distance to indoor units	20,00(m)	12,10(m)
Height difference between indoor and outdoor unit(ODU up)	30,00(m)	0,00(m)
Height difference between indoor and outdoor unit(ODU down)	20,00(m)	0,00(m)

Height difference between indoor units	10,00(m)	0,00(m)
Combination ratio	45-130%	128,00%
IDU quantity	9	4

4.4.2 Correction Factors

Item	Correction factor
Altitude (indoor unit)	1,000
Altitude (outdoor unit)	1,000
Piping (cooling)	0,996
Piping (heating)	1,000
Defrost (heating)	1,000

4.4.3 Piping Details Table

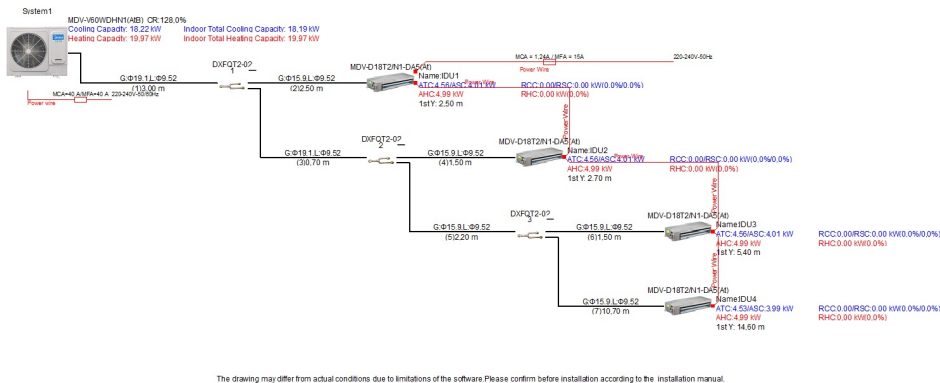
No.	Length(m)	Piping diameter
(1)	3,00	Φ19.1/Φ9.52
(2)	2,50	Φ15.9/Φ9.52
(3)	0,70	Φ19.1/Φ9.52
(4)	1,50	Φ15.9/Φ9.52
(5)	2,20	Φ15.9/Φ9.52
(6)	1,50	Φ15.9/Φ9.52
(7)	10,70	Φ15.9/Φ9.52

4.4.4 Branch Joints Details Table

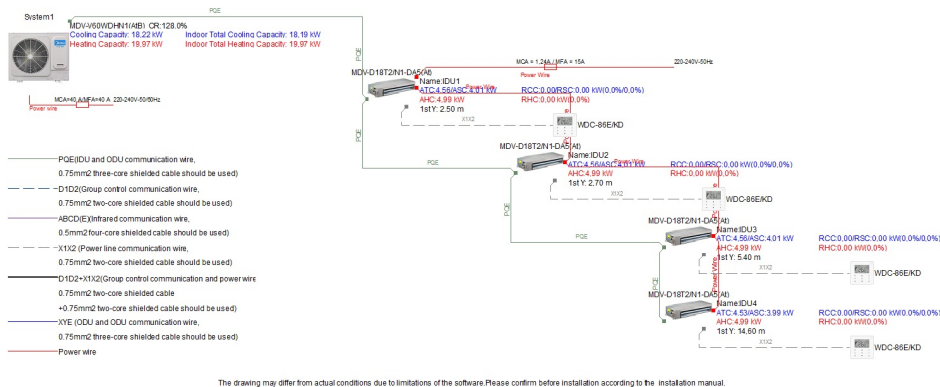
No.	Load(kW)	Model
(1)	22,4	DXFQT2-02
(2)	16,8	DXFQT2-02
(3)	11,2	DXFQT2-02

4.4.5 Reducer Details Table

4.5 Piping Diagrams (System1)



4.6 Wiring Diagrams (System1)



5. Centralized Control Solution

5.1 Centralized Controller List

The centralized control system of this project is full output regardless of whether the system is selected.